Cookies and Headers

FastAPI Masterclass

Header

- A **header** is a key-value pair attached to an HTTP request or response.
- Headers contain metadata, supplemental information.
- Headers do not persist; the client has to attach them to every request, and the server has to attach them to every response.

Headers in Network Tab

- Open the Network tab in Chrome, click on a request, and you’ll see the **Response** headers and the **Request** headers

▼Response headers

Access-Control-Allow-Origin	*
Alt-Svc	h3=":443"; ma=93600,h3-29=":443"; ma=93600
Cache-Control	max-age=300
Content-Encoding	gzip
Content-Length	427
Content-Type	application/json
Date	Mon, 13 Apr 2026 13:07:28 GMT
Etag	215cbca570dab055fe4d36557cf3b020

Sample Headers I

- **User-Agent** is a request header that includes the web browser and computer type.
  - **User-Agent** Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36
- A **user agent** is a computer program representing a person.

Sample Headers II

- **Content-Type** is a response/request header that communicates the type of a resource.
  - **Content-Type** application/json
  - **Content-Type** text/css
  - **Content-Type** text/plain; charset=utf-8

Cookies I

- A **cookie** is a key-value pair attached to a request but it has a special relationship with the browser.
- Cookies are often used for storing data like user preferences: theme, language, currency, etc.
- HTTP is a stateless protocol (a request doesn't know about previous requests). The browser attaches cookies on each request to provide the server with context.

Cookies II

- First, the browser makes a request to an endpoint.
- The server instructs the browser to store a cookie. A cookie has a name and a value.
- From that point on, the browser automatically attaches the cookie to every request to the same domain (same website like **airbnb.com** or **localhost**)

Cookies III

- Unlike headers, there is no manual configuration required on every request for cookies.
- Cookies persist across future requests to the same domain until they expire or are deleted.
- We can think of a cookie as an "automated header". Technically, a cookie *is* a header behind the scenes. Look for **Cookie** in the **Network** tab headers.

Cookie Security

- The browser will not share cookies across different domains for security reasons.
- For example, cookies set on **airbnb.com** will not be sent on requests to **linkedin.com** (different domain).
- You might have seen the “Accept cookies” banners on websites; they are asking for permission to store these cookies in your browser.